

Harmolight: Multimodal-Guided Diffusion Model for Composite Image Lighting Adaptation via Curriculum Lighting Learning

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Figure 1: Visual results of our method for composite image lighting adaptation. Given a foreground subject and a target background, our method relights the composited foreground to match the illumination of the new scene. The examples show that our method can handle both colorful lighting conditions, as in (a), and natural lighting conditions, as in (b).

Abstract

Composite image lighting adaptation aims to adjust the lighting of a composited foreground so that it harmonizes with the illumination of a target background. Recent diffusion models have significantly advanced relighting tasks by leveraging large-scale relighting pairs. However, existing methods often struggle to maintain both content consistency and lighting consistency in composite images. Moreover, the construction of relighting datasets and the corresponding training paradigm for lighting adaptation remain underexplored. To address these challenges, we propose HarmoLight, a multimodal-guided diffusion model for composite image lighting adaptation. Specifically, our model is guided by decoupled content and lighting prompts through a decoupled cross-attention mechanism, enabling more effective control over foreground appearance and illumination adaptation. To derive appropriate lighting guidance, we further introduce a lighting representation distillation strategy that transfers lighting encoding knowledge from a large pre-trained model. In addition, to better facilitate the learning of diverse lighting transformations, we employ a curriculum lighting learning strategy over

our specially designed datasets, where training samples are organized according to their lighting adaptation difficulty. Extensive quantitative and qualitative experiments on both synthetic datasets and real-world images demonstrate that our method consistently outperforms existing state-of-the-art approaches.

CCS Concepts

• Computing methodologies → Computational photography.

Keywords

Image relighting, Image harmonization, Diffusion model

1 Introduction

Composite image lighting adaptation aims to adjust the illumination of a composited foreground so that it is consistent with the lighting condition of a target background, thereby producing a realistic and harmonious composite image. This task is of great importance in photo editing, virtual content creation, and artistic design. In practical applications, manually adapting foreground

illumination with professional software, *e.g.*, Photoshop or Lightroom, is often labor-intensive and requires substantial expertise. Therefore, learning-based relighting methods have attracted increasing attention as an efficient alternative. However, composite image lighting adaptation remains challenging, as it requires the model to simultaneously preserve foreground content fidelity while achieving accurate lighting consistency with the background.

Early relighting studies [59, 76] typically adopt simple end-to-end neural networks, *e.g.*, convolutional neural networks (CNNs), trained on large-scale relighting datasets. Although these methods demonstrate the feasibility of data-driven relighting, their limited model capacity and dataset bias often restrict their generalization ability across diverse scenes and illumination conditions. Other decomposition-and-render approaches [33, 47, 49, 65] explicitly decompose intermediate physical components, such as albedo, depth, and surface normals, and then synthesize the relighted results through a rendering module conditioned on geometry and target illumination. While physically interpretable, these multi-stage pipelines are sensitive to estimation errors in the decomposition stage, which can propagate to the rendering stage and lead to visual artifacts and detail degradation. Meanwhile, harmonization-based methods mainly focus on adjusting the global color and tone of the foreground to better match the background, but often overlook more complex illumination effects, such as directional lighting, shading variation, and cast shadows.

More recently, diffusion models have shown remarkable success in image generation and relighting, benefiting from their strong generative capability and improved generalization across diverse visual scenarios. Nevertheless, directly adapting large-scale diffusion models to composite image lighting adaptation is still far from trivial. First, existing methods often struggle to maintain both content consistency and lighting consistency in composite images, especially when the target background exhibits complex illumination patterns. Second, the construction of relighting datasets and the corresponding training paradigm for lighting adaptation remain underexplored. Third, although conditional guidance plays a crucial role in diffusion models, how to inject complementary conditions that explicitly disentangle foreground semantics from background lighting has not been sufficiently investigated.

To address these challenges, we propose **HarmoLight**, a multimodal-guided diffusion model for composite image lighting adaptation. Our key idea is to decouple the guidance signals for content preservation and lighting adaptation, so that the model can better control foreground appearance and illumination transfer simultaneously. Specifically, we introduce decoupled content and lighting prompts and integrate them into the diffusion model through a decoupled cross-attention mechanism. To obtain reliable lighting guidance, we further introduce a lighting representation distillation strategy. Prior relighting studies conditioned on environment maps often rely on high-dynamic-range (HDR) environment lighting for precise illumination control. Inspired by this observation, we design a lightweight lighting encoder to extract aligned lighting representations from the target background image. Instead of directly depending on computationally expensive large-scale lighting estimation models, we distill lighting encoding knowledge from a powerful pre-trained model into the proposed encoder, thereby obtaining informative lighting embeddings with much lower computational cost.

Beyond model design, we also introduce a dedicated data construction pipeline and a curriculum lighting learning strategy to improve training effectiveness. Specifically, we first collect large-scale unpaired images from multiple sources and construct relighting pairs through three types of lighting transformation operators, namely filtered, delighted, and relighted. These transformations naturally exhibit different levels of adaptation difficulty. Based on this observation, we propose a two-stage curriculum lighting learning scheme. In the first stage, the model is trained progressively from easier to harder transformation types, allowing it to gradually acquire the ability to handle increasingly complex lighting changes. In the second stage, we further rank training samples according to their learning difficulty estimated by loss values, and continue training from easier samples to harder ones to enhance generalization across diverse real-world scenarios.

Our contributions can be summarized as follows:

- We introduce a multimodal-guided diffusion model with decoupled cross-attention, incorporating disentangled content and lighting guidance. Additionally, we introduce a lighting representation distillation strategy to derive a comprehensive lighting embedding.
- We introduce a two-stage curriculum learning strategy on a well-designed relighting dataset, which is constructed from a scalable data construction strategy and comprises data samples with different learning difficulty.
- Quantitative and qualitative experiments validate the effectiveness of our method across synthetic and real-world scenarios, outperforming state-of-the-art relighting and harmonization approaches.

2 Related Work

2.1 Single Image Relighting

Traditional relighting methods rely on light-stage devices [16] to capture portraits under multiple lighting conditions using controlled flashes. By estimating the reflection function at each pixel, these methods can synthesize images under different high-dynamic-range (HDR) lighting environment maps (envmaps). However, their practical applicability is limited by the difficulty and cost of acquiring such specialized equipment, which restricts their broader use. With the development of deep learning, recent studies [53, 56, 59, 76] have leveraged large-scale relighting datasets to train convolutional neural networks for modeling lighting variations. Although these methods have demonstrated promising performance, they are still constrained by dataset quality and the limited generalization ability of relatively small networks, which may lead to overfitting and reduced robustness in unseen scenarios. Other approaches [33, 49] incorporate physical lighting models through intrinsic decomposition, followed by re-rendering and compositing. While these methods are more physically grounded, they are susceptible to error accumulation across stages, which can ultimately degrade the quality of the final results. In contrast, we propose a diffusion-based relighting model built upon Stable Diffusion [55]. Owing to its generative nature, the model is more tolerant to imperfect conditions, and its strong prior learned from large-scale data endows it with better generalization across diverse scenes and lighting environments.

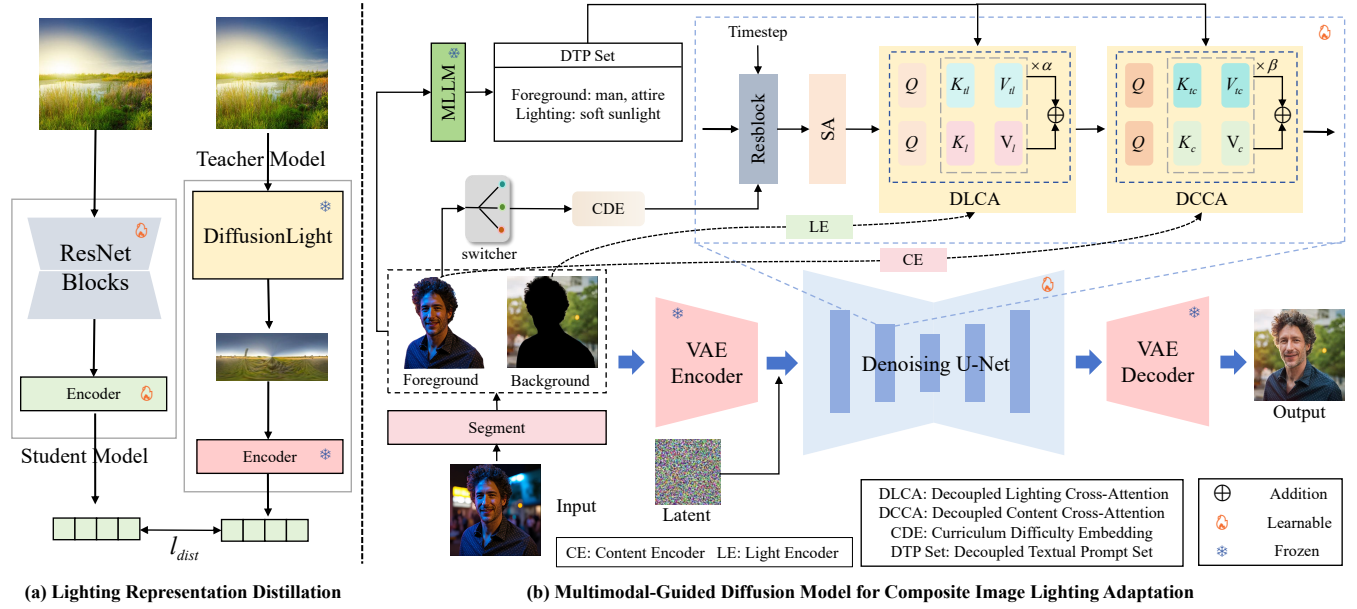


Figure 2: Overview of HarmoLight, our multimodal-guided diffusion model for composite image lighting adaptation. We first distill a lighting encoder (LE) from a large lighting estimation teacher model through lighting representation distillation, as shown in (a). Then, as illustrated in (b), our multimodal-guided relighting diffusion model integrates decoupled content prompts and lighting hint guidance to perform lighting adaptation. Moreover, we incorporate a curriculum difficulty embedding (CDE) to enable the model to handle diverse forms of lighting adjustment.

2.2 Diffusion Model for Relighting

Diffusion models have achieved remarkable progress across a wide range of vision tasks. Recently, several studies have begun to apply diffusion models to relighting problems [2–4, 11, 19, 20, 25, 36–41, 43, 44, 50, 63, 66, 70, 74, 77], including scene relighting [34, 73], object relighting [28, 69], and portrait relighting [6, 8, 21, 26]. However, the lighting conditions adopted in these methods, such as HDR environment maps or spherical harmonics (SH) coefficients, are often more difficult to obtain than ordinary images that already contain rich lighting cues. In contrast, our work focuses on background-controlled lighting adaptation, where the target background itself provides the illumination guidance for relighting. Among the most relevant works, IC-Light [71] and Relightful Harmonization [54] perform background-based relighting by fine-tuning diffusion models. Nevertheless, these methods have not fully explored the use of more effective textual guidance or more stable training strategies for learning diverse lighting transformations. Inspired by these advances, we propose a multimodal-guided diffusion model that incorporates decoupled textual prompts and lighting guidance for composite image lighting adaptation. In addition, we introduce a progressive lighting-aware curriculum training strategy to facilitate more effective and stable learning of complex lighting variations.

2.3 Single Image Harmonization

Image harmonization aims to composite a foreground into a target background and generate a visually coherent result. Existing harmonization methods [9, 13–15, 22–24, 27, 32, 42, 45, 60, 61, 67] mainly focus on global color and tone adjustment through end-to-end neural networks. Although effective for improving overall appearance

compatibility, they often overlook detailed lighting cues in the target background and thus fail to achieve consistent illumination adaptation for the foreground. Among lighting-aware methods, Relightful Harmonization [54] and Intrinsic Harmonization [5] are the most relevant to our task. However, Relightful Harmonization mainly targets portrait images, while Intrinsic Harmonization relies on offline pre-trained intrinsic decomposition models, which increases pipeline complexity and may introduce cumulative estimation errors. By contrast, our method leverages large-scale data and diffusion-model priors to learn composite image lighting adaptation more effectively, enabling better illumination consistency and broader generalization to diverse natural scenes.

3 Method

The task definition of relightful harmonization is to change the foreground lighting to align with the given new background image. We present a decoupled prompt diffusion model, termed HarmoLight, for our task. Its training dataset, model architecture, and training strategy are detailed in Sections 3.2, 3.3, and 3.4, respectively. Additionally, we introduce our decoupled prompt representation for relighting in Section 3.1 and Section 3.3.

3.1 Lighting Representation Distillation

The embedded lighting representation of the target lighting background provides the lighting guidance cues for diffusion-based relighting. Given a target lighting background image, we derive the lighting embedding in a more efficient method by a lighting representation distillation strategy.

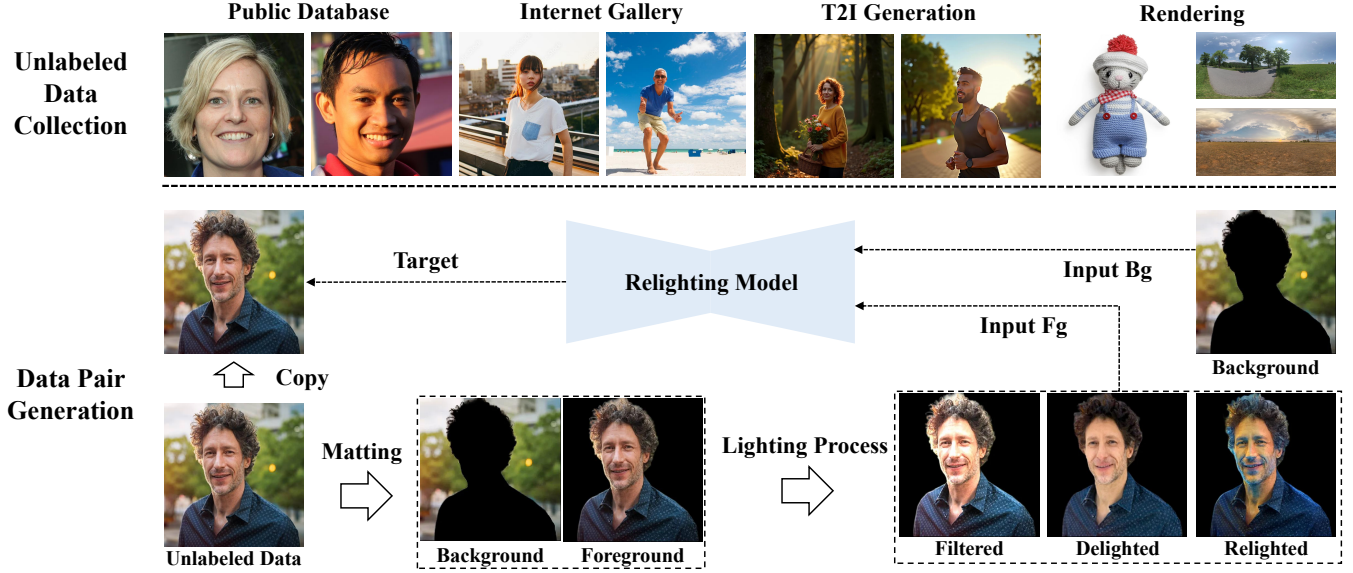


Figure 3: The pipeline for constructing relighting data pairs in a self-supervised manner. We first collect a large amount of unlabeled data from multiple sources. Then, we employ existing lighting adjustment models and operators to alter the foreground illumination of real images in the unlabeled database, thereby creating relighting pairs with different lighting conditions. In addition, we incorporate rendered data as a supplementary source, where objects are rendered under different high-dynamic-range (HDR) environment map lighting conditions.

Motivation. Diffusionlight [51] and Diffusionlight-Turbo [10] are large lighting estimation models that generate HDR environment maps from input images via multi-step and single-step diffusion iterations, respectively. By fine-tuning on SDXL [52] with a tailored dataset, these models inherently learn a lighting representation of the environmental lighting from a single image.

However, the large memory usage and slow inference speed make it impractical to utilize these models to generate lighting environment maps online for our relighting harmonization task. A straightforward idea is to distill the lighting representation capacity from these models into a student model. Nevertheless, predicting an intricate HDR environment map is a challenging task for a simple student model.

Additionally, we only need a compact lighting representation embedding to guide the diffusion relighting process, rather than reconstructing the exact appearance of the environment map. Therefore, we design a student model to directly estimate the lighting representation of the background image under the supervision of the teacher model.

Distillation Procedure. To distill its lighting representation capacity, we design a small student model consisting of a convolution-based projector network and an image encoder. Specifically, the projector has a U-shaped architecture, and each layer consists of ResNet blocks to derive a coarse environment map, while the image encoder is initialized from the CLIP image encoder to embed the lighting. In contrast, the teacher model consists of a single-pass process of Diffusionlight-Turbo to predict the HDR lighting environment map by fused the outputs of Diffusionlight-Turbo with multiple exposures. The derived HDR envmaps are transformed into the LDR images following Neural Gaffer [28] and fed into a

CLIP image encoder to embed the environment map into a lighting representation.

In the training process, we freeze the teacher model and train the student model using a distillation MSE loss $\mathcal{L}$ as follows:

$$\mathcal{L} = l_{\text{diss}}(f_{\text{student}}, f_{\text{teacher}}), \quad (1)$$

where f_{student} and f_{teacher} are encoded lighting embeddings from the student model and teacher model, respectively.

3.2 Curriculum-based Relighting Dataset

To construct our relighting dataset, we comprehensively utilize the paired rendering data and numerous unpaired data. For the unpaired data, we collect from multiple sources, including in-the-wild data and AIGC data, and then we design a method for constructing relighting data pairs.

Paired Rendering Data. We utilize over 10K 3D models from the Objaverse dataset [17] and more than 700 HDR environment maps downloaded from the internet. Each environment map was enhanced by randomly rotating it at various angles. Using these 3D models and environment maps, we rendered images of each 3D object under different backgrounds using the Cycles renderer in Blender [12]. For each 3D model, we rendered four different viewpoints and five distinct lighting conditions, resulting in over 200K images with a resolution of 512×512.

Unlabeled Data Collection. We collect a large number of unlabeled images, including in-the-wild data and AIGC data. We collect a large number of in-the-wild images from the internet and existing public datasets. Specifically, we use the publicly available portrait dataset FFHQ [30], which contains over 50K images at a resolution of 512×512. Additionally, we download over 20K complementary

in-the-wild images from the gallery websites to enrich our dataset. In addition, we collect a large number of AIGC images to enrich our data. Specifically, we first utilize ChatGPT [48] to generate text descriptions of various types of foregrounds, backgrounds, and lighting conditions. Then, we use ChatGPT to randomly combine these foreground, background, and lighting descriptions into over 10K complete image descriptions. Finally, we use FLUX.1-dev [35] to generate over 10K corresponding images according to the text descriptions.

Relighting Pair Data Formulation. We construct our training pairs as shown in Fig.3. Specifically, given an original image, we first separate the foreground and the background using a pre-trained segmentation model, rmbg2.0 [75]. Then, we apply three lighting processing operators to the separated foreground to generate foreground images with varying lighting conditions: filtered, delighted, and relighted. Finally, our objective is to align the lighting of the relit foreground with that of the original background.

Decoupled Textual Prompt Generation. We leverage the visual comprehension capabilities of the MLLM GPT-4o [1] to generate decoupled text prompts that describe image content and lighting conditions separately. To achieve this, we design a prompt template: "Please analyze the image and generate descriptive tags in JSON format, including the foreground, background, and lighting." This template facilitates dialogue with the MLLM to obtain decoupled text prompts.

3.3 Multimodal-Guided Diffusion Model

As shown in Figure 2, given an input image I under the unknown lighting condition, we segment the foreground I_{fg} from I and derive the foreground mask I_{mask} . Given the target lighting background I_{bg} , we input I_{fg} and I_{bg} into our model $f(\cdot; \theta)$ to predict the relighting result by iteratively denoising. To enable content preservation and lighting awareness, we inject content guidance C_c and lighting representation C_l conditions to guide the diffusion process, where C_c and C_l are constructed from the input I_{fg} and I_{bg} .

To adapt the pre-trained text-to-image Stable Diffusion model [55] to our relightful harmonization task, we modify the vanilla Stable Diffusion architecture (*i.e.*, SDv1.5). The original SDv1.5 consists of a VAE encoder, a decoder, a denoising U-Net, and optional text and image encoders. We increase the input channel size of the denoising U-Net to accommodate both the foreground and background conditions. Additionally, we incorporate the curriculum difficulty embedding (CDE) and the decoupled attention mechanism to adapt our model for curriculum learning and decoupled prompt conditions.

Specifically, we derive the textual descriptions of the foreground content t_c and background lighting t_l by utilizing an off-line multimodal large language model (MLLM) GPT-4o [1]. Additionally, we derive the foreground embedding e_c from input I_{fg} by the CLIP image encoder and lighting representation e_l from I_{bg} by the lighting representation encoder.

Curriculum Difficulty Embedding. Furthermore, we slightly modify the architecture of the denoising U-Net. We project the index i of the relighting curriculum (*i.e.*, relighted, delighted, or filtered) into an embedding CDE, and inject it into the resblock of the U-Net via a method similar to the timestep. We obtain the

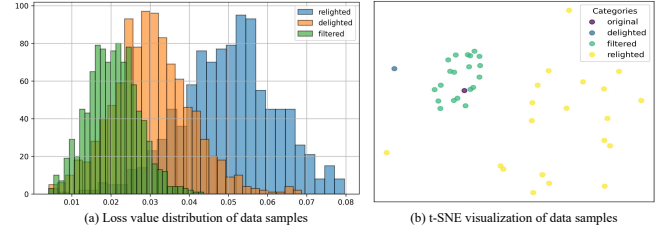


Figure 4: t-SNE visualization of the distribution of different relighting data pairs, together with the distribution of randomly sampled loss values, illustrating that our constructed curriculum training pairs cover diverse levels of learning difficulty.

curriculum difficulty embedding as follows:

$$I = \text{cat}[\sin(i), \cos(i)], \text{CDE} = \text{MLP}(I), \quad (2)$$

Decoupled Attention Mechanism. To guide the diffusion relighting process, we inject lighting guidance and content guidance conditions via a cross-attention mechanism. Mixing all these conditions together increases the learning difficulty of the relighting model. Motivated by IP-Adapter [68], we design decoupled cross-attention to process different conditions, *i.e.*, decoupled content cross-attention (DCCA) and decoupled lighting cross-attention (DLCA) to process content guidance C_c and lighting representation C_l , respectively. The main process of the denoising U-Net is as follows:

$$Y = \text{SA}(X), \quad (3)$$

$$Y = \text{DCCA}(Y), \quad (4)$$

$$Y = \text{DLCA}(Y). \quad (5)$$

where SA is the self-attention mechanism in the denoising U-Net.

The vanilla cross-attention CA(X,Y) mechanism is as follows:

$$Q = \text{Linear}(X), \quad (6)$$

$$K = \text{Linear}(Y), \quad (7)$$

$$V = \text{Linear}(Y), \quad (8)$$

$$\text{CA} = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \cdot V \quad (9)$$

To provide lighting awareness guidance, we use lighting representation embedding c_l and lighting textual representation t_l . Then, the DLCA is as follows:

$$\text{DLCA}(X, c_l, t_l) = \text{Attention}(X, c_l) + \alpha \cdot \text{Attention}(X, t_l), \quad (10)$$

where α is a learnable parameter. To provide content detail guidance, we use foreground content embedding c_c and foreground textual representation t_c . Then, the DCCA is as follows:

$$\text{DCCA}(X, c_c, t_c) = \text{Attention}(X, c_c) + \beta \cdot \text{Attention}(X, t_c), \quad (11)$$

where β is a learnable parameter.

In the training process, we first encode the original foreground image I_{fg} , the target background image I_{bg} , and the target harmonious image I_{gt} through the VAE encoder. Then, the target latent is added noise $\epsilon \in \mathbb{R}^{H' \times W' \times C}$ at random timestep t to turn into a noisy latent z_t by $z_t = \sqrt{\alpha_t} z_0 + \sqrt{1 - \alpha_t} \epsilon$ and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. Then the encoded foreground, background, and the noisy latent features are concatenated along the channel dimension. This concatenated

Method	Venue	Synthetic Test Set			Rendered Test Set			In-the-wild Test Set		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	CLIPQA↑	MUSIQ↑	NIQE↓
Harmonizer	ECCV2022	16.743	0.773	0.235	22.544	0.893	0.088	<u>0.749</u>	<u>76.562</u>	3.039
PCT-Net	CVPR2023	16.932	0.770	0.217	23.854	0.911	0.078	0.732	75.988	2.942
AICT	NeurIPS2024	17.782	0.779	0.202	23.900	0.908	0.079	0.745	75.034	<u>3.020</u>
IC-Light	ICLR2025	18.089	0.728	0.235	18.865	0.868	0.109	0.716	74.181	3.069
LBM	ICCV2025	<u>18.363</u>	<u>0.787</u>	<u>0.176</u>	<u>24.129</u>	0.896	<u>0.076</u>	0.695	73.525	4.079
Ours	-	23.273	0.819	0.102	24.782	<u>0.906</u>	0.072	0.789	76.574	3.110

Table 1: Quantitative comparison with other state-of-the-art methods on synthetic, rendered, and in-the-wild test sets. The best and second-best results are highlighted in bold and underlined, respectively.

representation is then fed into the denoising U-Net with timestep t , content guidance C_c , lighting representation C_l , and curriculum difficulty embedding CDE to predict the noise as follows:

$$\mathcal{L} = \mathbb{E}_{z_0, \epsilon, t} \|\epsilon - \epsilon_\theta(z_t, t, \text{CDE}, C_c, C_l)\|_2^2. \quad (12)$$

where ϵ denotes VAE image encoder.

3.4 Curriculum Lighting Learning

To achieve effective and stable training, we introduce a curriculum lighting learning strategy, progressively training on easier to more difficult relighting samples. This approach facilitates the learning of relighting. Specifically, we construct three different lighting adjusting operators in Section 3.2: filtering, delighting, and relighting. These operators implicitly learn to: (a) regress random lighting filter errors to match the same lighting under the same background, (b) perceive the background lighting and interact with the known geometry, and (c) perceive the geometry decoupled from shading and interact with the perceived lighting.

Figure 4 visualizes the learning difficulty of the three lighting adjusting operators by evaluating the training loss for each sample. The loss values increase from 'filtered' to 'delighted' to 'relighted,' indicating a progression from easier to harder samples. We also cluster the distributions of the three sample types as shown in Fig. 4 and find that the distance from the target increases in the same order. The three lighting adjusting operators are described as follows:

- (1) **Filtering operator.** We apply a combination of color enhancement, contrast enhancement, and brightness enhancement with random ratios between 0.7 and 1.3 to the original image, simulating an image filtering operator.
- (2) **Delighting operator.** Albedo preserves only the geometric content and base color of the image, independent of the lighting. We extract the albedo from the image to simulate the delighting operator, using the pre-trained albedo decomposition method StableDelight [58].
- (3) **Relighting operator.** The pre-trained relighting method IC-Light [71] can relight the foreground while preserving the intrinsic geometry. We use IC-Light to relight the original foreground with a new lighting background, simulating the relighting operator.

By employing these three operators, we construct relighting data pairs of varying difficulty.

Curriculum Learning Process. We design a two-stage progressive curriculum training procedure to facilitate lighting learning,

including the first stage heuristic-based curriculum scheduling CL_1 and the second stage automated curriculum scheduling CL_2 .

In the first stage, based on the observation that three types of relighting data pairs have different learning difficulties by loss calculation in preliminary experiments and clustering analysis, we train the model from simple to hard data samples, *i.e.*, filtered data, delighted data, and relighted data. However, after some iterations, the learning difficulty of data samples varies, and the model has acquired partial knowledge about different types of lighting transformations. To focus on intricate relighting translation and generalize better to more scenarios, we train our model only on the relighted data. The two-stage process is as follows:

$$P(t) = \begin{cases} \text{CL}_1, & \text{if } t < T \\ \text{CL}_2, & \text{if } t \geq T \end{cases} \quad (13)$$

where t denotes the training iteration, and T is a manually chosen hyperparameter representing the threshold that divides the two stages. In the second stage, we pre-compute the loss values by predicting the noise in the data samples within the same batch setting. Then, according to the descending order of these loss values, we progressively train on the samples.

4 Experiments

4.1 Experimental Settings

Dataset. We train our relighting model on the training split of our constructed dataset, as described in Section 3.2, which contains approximately 550K data pairs and consists of both rendered and synthetic data. We evaluate our model on the test split of our constructed dataset as well as on in-the-wild images collected from the Internet, with each test set containing 500 data pairs.

Metrics. For the synthetic and rendered test sets with ground-truth labels, we evaluate relighting performance using reference-based metrics, including PSNR, SSIM [64], and LPIPS [72]. For the in-the-wild test set, which does not provide ground-truth labels, we evaluate the relighting results using non-reference metrics, including CLIPQA [62], MUSIQ [31], and NIQE [46], to assess the perceptual quality and naturalness of the generated relighting results.

Compared Methods. We compare our method with several state-of-the-art approaches, including the relighting methods IC-Light [71] and LBM [7], as well as the harmonization methods Harmonizer [32], PCT-Net [22], and AICT [45].

Implementation details. Our method is implemented based on Stable Diffusion v1.5 [55]. We modify the input channel dimension of the first convolution layer in the denoising U-Net to incorporate both foreground and background conditions. We use the Adam

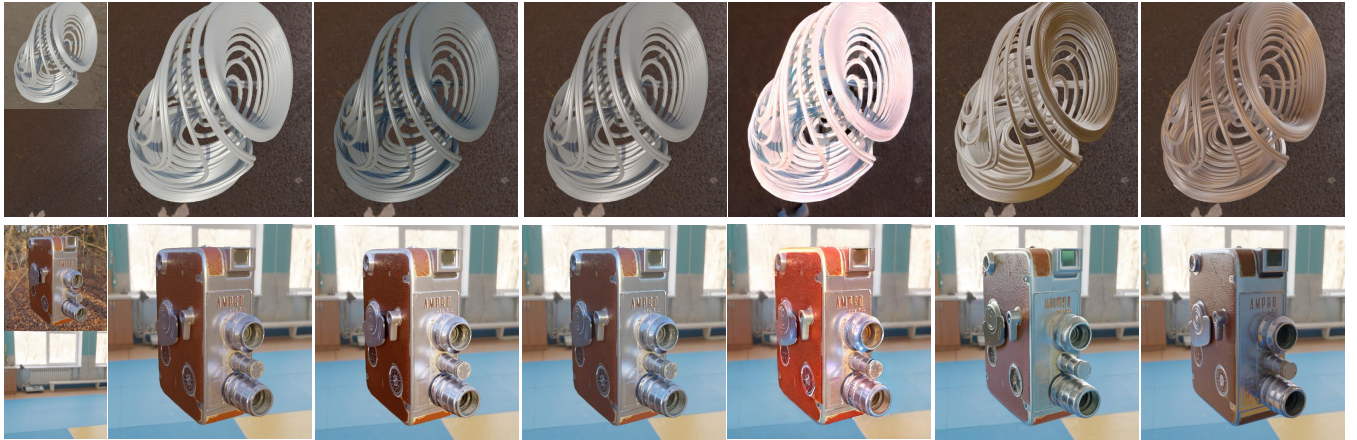


Figure 5: Qualitative comparisons with other methods on the test set of our rendered object dataset. Our method produces colors that are most consistent with the ground-truth images.

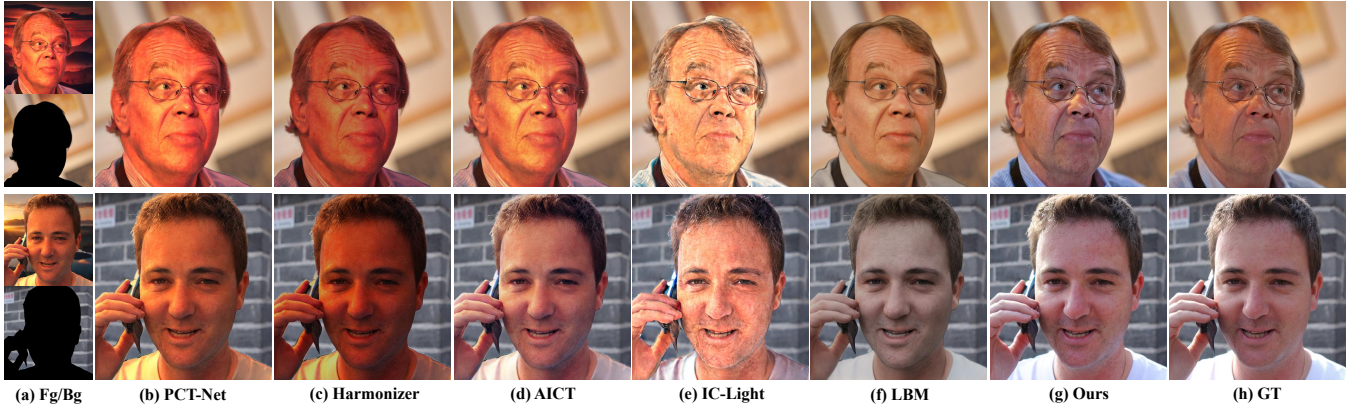


Figure 6: Qualitative comparisons with other methods on the test set of our synthetic dataset. Our results achieve the best consistency with the target images in both lighting and content.

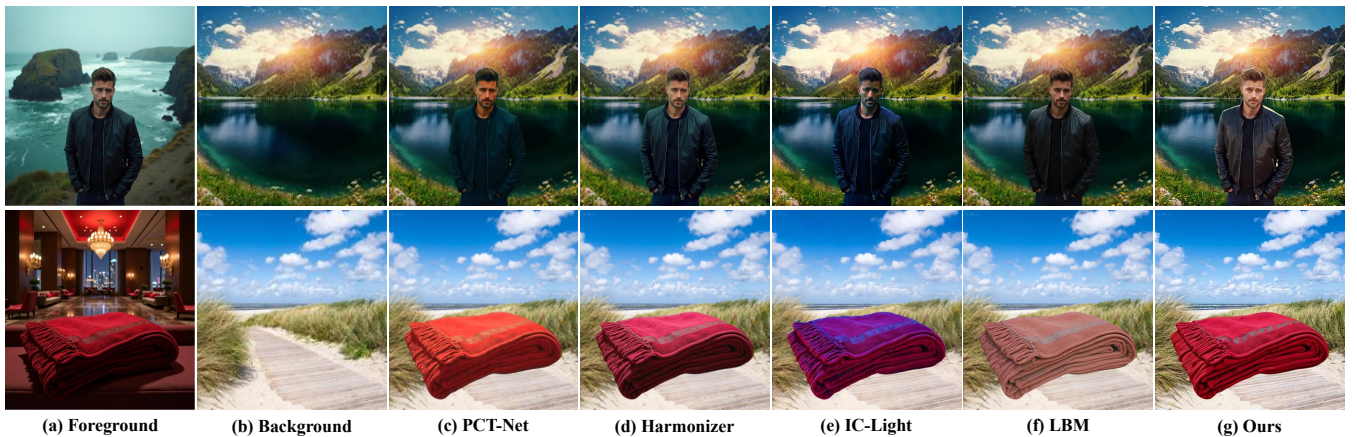


Figure 7: Qualitative comparisons with other state-of-the-art methods on in-the-wild data. Our results exhibit more pronounced lighting adaptation and better match the background images, including both local changes in lighting direction and global adjustments in overall brightness.

optimizer [18] for training, with a learning rate of $5e-5$. Training is conducted on NVIDIA A100 80GB GPUs. All training images are resized to a resolution of 512×512 , and the model is trained with a two-stage curriculum learning strategy. In the first stage, we train the model on retouched data pairs for approximately 1K steps, followed by delighted data pairs for approximately 2K steps, and then on relighted data pairs for one epoch. In the second stage, we further train the model on relighted data pairs to complete the curriculum learning process. In total, the model is trained for approximately 20K steps. During inference, we adopt the DDIM sampler [57] with 50 denoising steps. We set the classifier-free guidance (CFG) scale to 2.0 during inference. Finally, we use alpha blending [29, 49] to composite the relighted foreground with the target background and obtain the final output.

4.2 Comparisons with State-of-the-Art Methods

Quantitative Comparisons. The quantitative results are presented in Table 1. On the synthetic test set, our method outperforms state-of-the-art image relighting and image harmonization methods in terms of PSNR, SSIM, and LPIPS. On the rendered test set, our method achieves the best performance on PSNR and LPIPS, and ranks second on SSIM. These results demonstrate that our method is highly effective in both preserving foreground content consistency and learning accurate lighting transformations. For the in-the-wild test set, where ground-truth relighted images are unavailable, we evaluate the realism of the relighting results using no-reference metrics. Our method achieves the best performance on both CLIPQA and MUSIQ, further validating its superiority in real-world scenarios.

Qualitative Comparisons. The qualitative results on the rendered test set are presented in Figure 5. Our method produces colors that are most consistent with the ground-truth images. The qualitative results on the synthetic test set are presented in Figure 6. Our results achieve the best consistency with the target images in terms of both lighting and content preservation. The qualitative results on the in-the-wild test set are shown in Figure 7. We observe that image harmonization methods tend to produce outputs that remain close to the original composite image, mainly adjusting the global color or tone of the foreground while failing to capture detailed changes in illumination and shadows. In addition, advanced relighting methods such as IC-Light [71] and LBM [7] often exhibit noticeable color shifts that are less compatible with the target background.

Method	PSNR↑	SSIM↑	LPIPS↓
Base	21.119	0.803	0.126
(a) w/ single cross-attention	22.362	0.808	0.113
(b) w/o textual prompts	23.315	0.813	0.124
(c) w/o lighting embedding	23.102	0.812	0.107
(d) w/o curriculum learning	22.720	0.805	0.117
Ours	23.273	0.819	0.102

Table 2: Quantitative ablation results of our condition guidance mechanism, model design, and training strategy.

4.3 Ablation Study

We define the diffusion model with an empty textual prompt and null lighting embedding, a single cross-attention mechanism, and a

Method	PSNR↑	SSIM↑	LPIPS↓
Ours + SD v1.5	23.273	0.819	0.102
Ours + SD 2.1	22.897	0.804	0.113
Ours + SDXL	23.956	0.824	0.097

Table 3: Quantitative comparison of our method with different diffusion backbones.

basic training paradigm as the base model. Additionally, we design several ablation studies to validate the effect of our key design, including a) our model using a single cross-attention mechanism with both text and lighting guidance, b) our model without textual prompt guidance, c) our model without lighting embedding guidance, and d) our model trained without using the curriculum lighting learning strategy. The results of these models are shown in Table 2.

Multiple Condition Guidance. We study the effectiveness of textual prompt guidance and lighting embedding guidance by comparing models b) and c) against the base model and our full method. The quantitative results indicate that textual prompts slightly reduce pixel-wise fidelity while improving perceptual quality, whereas lighting embeddings provide more explicit illumination cues and thus help the model learn lighting transformations more effectively. **Decoupled Cross-Attention Mechanism.** We investigate the effectiveness of the decoupled cross-attention mechanism by comparing model a) with the base model and the full model. The quantitative results show that decoupled cross-attention enables more effective integration of different guidance signals, thereby facilitating relighting and improving overall performance.

Curriculum Lighting Learning. We further evaluate the effectiveness of the curriculum lighting learning strategy by comparing model d) with the base model and the full model. The quantitative results demonstrate that curriculum lighting learning improves model performance, confirming its benefit for learning diverse and increasingly complex lighting transformations.

Scaling Up with Different Backbone Models. We further evaluate the scalability of our method by adopting different diffusion backbones, including SD 2.1 and SDXL. As shown in Table 3, stronger backbone model, *i.e.*, SDXL, improves the performance of our method.

5 Conclusion

In this paper, we presented **HarmoLight**, a multimodal-guided diffusion model for composite image lighting adaptation. By introducing decoupled content and lighting prompts together with a decoupled cross-attention mechanism, our model is able to better preserve foreground content while adapting its illumination to match the target background. To provide more effective lighting guidance, we further proposed a lighting representation distillation strategy that transfers lighting encoding knowledge from a large pre-trained model to a lightweight lighting encoder. In addition, we developed a curriculum lighting learning strategy on a specially designed dataset, enabling the model to progressively learn increasingly challenging lighting transformations. Extensive experiments on both synthetic datasets and real-world images demonstrate that our method consistently achieves state-of-the-art performance, validating its effectiveness for realistic and harmonious composite image lighting adaptation.

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